Chu-Yin Weng

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EDUCATION

Cornell University, College of Engineering Ithaca, NY
B.S. in Computer Science | Minor in Information Science | Dean's List All Semesters, Cum Laude May 2026
Relevant Coursework: Algorithms, Object-Oriented Programming and Data Structures, Distributed Systems Programming, Operating Systems, Discrete Structures, Machine Learning, Database Systems, Robotics, Functional Programming, Linear Algebra

SKILLS

Languages: Ruby, Java, C++, Python, TypeScript, JavaScript, SQL, OCaml
Frameworks & Testing: Ruby on Rails, Sidekiq, RSpec, React, Next.js, Three.js, Flask, Apache Spark
Data & Infrastructure: AWS (Glue, Athena, S3, EventBridge, CloudWatch, CDK), PostgreSQL, Snowflake, Firebase, Supabase

EXPERIENCE

SWE, *Kikoff* August 2026 – Present
- Designed and developed an identity verification backend for onboarding workflows serving **250K+ users monthly**, leading technical discussions with vendor engineers to resolve integration requirements and implementation details.
- Designed multi-stage verification logic with Compliance to balance onboarding approval rates and customer risk, validating the integration through shadow runs, Snowflake analysis, and A/B testing before broader rollout.
- Diagnosed and resolved an onboarding issue affecting approximately **3,700 users** by analyzing customer event data, enabling applicants to complete additional identity checks and reducing manual review time for Customer Experience teams.
- **Halved operator time spent** on internal credit-reporting uploads by redesigning post-upload processing to run asynchronously and concurrently with Sidekiq, and improving the UI to clearly display upload status and results.
- Built a full-stack office-navigation platform integrating NFC stickers for mobile access to interactive maps and meeting-room search; partnered with Infrastructure and Information Security to enable employee access without a corporate VPN.

SDE Intern, *Amazon Web Services* May - August 2025
- Built and deployed a scalable, automated post-processing validation system using AWS Glue with Apache Spark, EventBridge, and CDK, processing **15K+ daily data records**, strengthening quality checks in production workflows serving **1M+** customers.
- Developed custom data validation logic including schema enforcement, null detection, and field-specific rules, increasing pipeline data integrity and reducing debugging time by **20%+** through early error detection.
- Reduced data quality–related on-call tickets by **25%**, improving pipeline reliability for 3+ downstream teams.

R&D Intern, *Industrial Technology Research Institute* June - August 2024
- Built a local RAG chatbot pipeline using pdfplumber for PDF extraction, SentenceTransformer embeddings, a FAISS vector store **(30K+ vectors)**, Llama3, LangChain, and Streamlit.
- Crawled and preprocessed **900+** disease-related documents from web and PDF sources, integrating a document retriever + LLM re-ranker to improve answer accuracy by **50%**.
- Evaluated chatbot accuracy using hospital-sourced workflows in order to improve reliability and contextual accuracy.

PROJECTS

Freelance Software Developer, ScholarSAT May - June 2026
- Developed a Next.js and TypeScript application for a tutoring-center client, supporting student score tracking, parent progress viewing, and administrative workflows with Firestore.
- Implemented role-based access with Firebase Auth and custom admin approval, alongside a question bank supporting manual entry and PDF/HTML ingestion.

Freelance Software Developer, QuickFi April - May 2026
- **Reduced transaction bottleneck detection time by 80%** by replacing manual data analysis with a natural language query interface, enabling teams to surface cycle time insights and performance outliers on demand.
- Designed an OpenAI API integration with tiered context injection, grounding responses in precomputed transaction metrics to support natural-language analysis.
- Engineered a Python ETL pipeline using pandas to process **1K+ transactions**, computing stage durations, aggregations, and IQR-based outlier detection across lender and dealer scopes.